Singha Junchan

Security Engineering: Cloud Security · Identity & Access (IAM) · AI-Agent Security · Governance & Risk
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web portfolio

Intro

As an ambitious, security-focused computer engineer, I see security as an enabler that lets a business move ahead with confidence. I seek out demanding challenges and meet them with discipline, as more than 1,000 hours of training for the national programming olympiad show. Research-grade depth in zero-knowledge proofs, post-quantum cryptography and zero-trust cloud access pairs with creative, hands-on engineering that starts from the root cause of each problem.

I am looking for a cloud security, IAM, AI security or governance role: an internship from January 2027 and a full-time role from May 2027.

Education

B.Eng. in Computer Engineering, specialisation in Cloud Computing & Cybersecurity — Aug 2023 – May 2027 (expected)
Sirindhorn International Institute of Technology (SIIT), Thammasat University — *GPA 3.67 / 4.00*

- Expected first-class honours; full scholarship holder (4-Year Continuing Academic Outstanding Student Program).
- **Relevant coursework:** Computer and Communication Security, Cyber Crimes and Digital Forensics, Data Structures and Algorithms, Computer Network and Internetworking, Cloud Computing, Ethical Hacking, Artificial Intelligence and Applications.

Experience

Information Security Intern – *Allianz Ayudhya, Bangkok* — Jun – Jul 2026

Every supplier relationship introduces risk, and third-party risk management (TPRM) exists to contain it, yet each review was largely manual. Mentored by the CISO, learned how TPRM operates within ISO 27001, NIST and the company's internal framework. Designed and built two AI agents that support the team rather than replace it: each does the routine first pass and a person verifies every output. Both are now in production use by the Information Security team and Business Owners.

- Supplier side: an agent drafts answers to partners' due-diligence questionnaires from internal policies and evidence, each backed by sources, a rationale and a confidence score, so reviewers begin from an evidence-based draft rather than a blank form; it is designed never to disclose internal data and suggests a fix for every gap it finds.
- Buyer side: an assistant guides Business Owners step by step through supplier assessments and validates inputs; it warns which parts need close scrutiny, a second check or a rethink, and what is at stake if key facts are missed, so owners learn to run assessments alone.

Tech Lead – *TU Alert, built for Thammasat University, launching soon* — 2026 – Present

TU Alert began at a Thammasat University hackathon that the team attended. The university adopted the idea and commissioned the team to build it, because campus had no channel to the office able to act on an incident.

Once launched, it should let anyone on campus report from the app they already use, alert the right office and give the university a record of every incident, response times and follow-through. Leads the team; owns architecture, security design and code.

- Capabilities: students report within LINE with a photo and a map pin (anonymously if they wish) and follow their case; reports reach the right office by category and campus zone, where staff get live updates, assign cases and see analytics.
- Security by design: protects against account takeover, access to other people's reports, spam, brute-force abuse, forged messages and malicious uploads; sensitive staff actions need stronger verification and are always recorded.
- Privacy and reliability: public map pins are blurred and consent, export and erasure are built in; notifications are retried.

Technologies: *TypeScript, Hono on Bun, Next.js, React, LINE LIFF, MariaDB, Redis.*

R&D Intern, AI Research Team – *Amity Solutions, Bangkok* — Jul – Aug 2024

Joined while large language models were still maturing, when every chatbot needed careful engineering to earn trust. Explored the newest GPT and Claude models and tested recent LLM research end to end, from question to sourced answer.

- Retrieval-augmented generation (RAG): built chatbots that answer from a knowledge base; prepared the source documents, tuned retrieval so each question reaches the right passages and explored document parsing and vector databases.
- Trust by design: answers stay grounded in the sources to avoid guessing, hallucination and leaks; tested on large data volumes.
- Evaluation: ran an LLM-judge benchmark of Thai banking and healthcare questions; answers were scored and corrections proposed.

Research

Eleven projects with Prof. Somchart Fugkeaw (SIIT), each starting from a genuine weakness in how organisations secure cloud systems and answering it with a rigorous design fast enough for business use.

FEDAC – *zero-trust access without the latency penalty* — First author · submitted, IEEE Trans. Dependable and Secure Computing

Federated sign-in proves identity, not permissions, so every request needs repeated checks, costly proofs or a blockchain call.

FEDAC decides once: it compiles the governance-checked decision into a reusable, service-scoped result, and a sidecar beside each service validates it locally, with no blockchain call.

- Decide once, check locally: decision-making is separated from enforcement, so the costly step runs once and every later request needs only a local check; assurance scales with risk across four levels, from OIDC sign-in to a zero-knowledge proof.
- Privacy and governance by design: service-scoped pseudonyms prevent identity correlation, and Merkle-committed policies on a permissioned ledger make rollback detectable; evaluated on AWS EKS against three re-implemented frameworks.

Technologies: *Go, Rust (arkworks, Groth16), OIDC/JWT, TOTP, IPFS, Solidity smart contracts, Kubernetes, AWS EKS.*

U-HAP – *one policy language for Kubernetes, compiled for speed* — Best Paper Award · Second author · IEEE JCSSE 2026

Kubernetes authorises requests with RBAC, yet organisations also need ACLs, attribute-based rules and explicit denies. Defined separately, the four models fragment checks, settle conflicts implicitly and slow decisions down.

U-HAP expresses all four in one policy graph, compiled in advance into per-resource lookup tables behind a Kubernetes webhook.

- A deny always prevails: a fixed order of evaluation (deny, ACL, RBAC, ABAC, default deny) means no allow rule can override an explicit deny; compilation, pruning and caching keep the cost of a decision independent of policy size.

- Evaluation: matched Open Policy Agent decision for decision, with faster policy updates in lab benchmarks; paper presented.
  Technologies: *Python, Kubernetes webhooks, Z3, OPA/Rego* | *doi.org/10.1109/JCSSE68839.2026.11597088*

ZKAudit, ZK-ALBA and PQZ-Audit – *senior project: private log audit* First author · ZKAudit accepted, KST 2027

Auditing cloud logs poses a dilemma: the provider that holds them could alter, delete or withhold records, yet the logs are sensitive and even who may audit must stay private. Existing schemes verify an auditor's right and the log's integrity separately, or funnel every proof through a single verifier.

This work joins the two, privately and at scale, in three steps: ZKAudit, ZK-ALBA and PQZ-Audit.

- ZKAudit proves in zero knowledge that an auditor holds a valid credential for the audit scope and binds that admission to the exact anchored log state, so a substituted log is rejected; sharded, batched verifiers serve many concurrent audits.
- ZK-ALBA turns the proof into a single-use capability bound to authorisation state, log, target and requester, so a replay is useless; auditors verify origin, inclusion and completeness of encrypted records without plaintext.
- PQZ-Audit lets auditors run checks over encrypted logs (windows, thresholds, totals) and verify the answer, staying secure against "harvest now, decrypt later"; t-of-n committees keep one authority from both authorising and recovering.
  Technologies: *Rust (arkworks, Groth16, Poseidon), Merkle trees, Fabric chaincode (Go), ML-KEM, ML-DSA.*

PQ-SCVA-MCK – *post-quantum trust across Kubernetes clusters* First author · submitted

Across Kubernetes clusters, an enforcer trusts the controller's answer: a compromised controller can drop a deny rule and still return a validly signed "allow", using signatures a quantum computer could one day forge.

PQ-SCVA-MCK verifies first: the enforcer checks each decision against the full committed rule set, and later checks stay light.

- Distinctive: proves the complete decision, uses post-quantum signatures and makes no blockchain call per request, where related schemes only authenticate records; the guarantee holds even if workloads and authorisation components are compromised.
- Security and evaluation: defends against omitted deny rules, stale proofs, rollback, forged capabilities and bypassed revocation; sustains higher throughput than three re-implemented schemes on AWS EKS with a live ledger.
  Technologies: *Go, ML-DSA (Cloudflare CIRCL), Hyperledger Fabric, Kubernetes, AWS EKS.*

SWARM-Fog – *self-healing orchestration for industrial IoT* First author · submitted

Industrial fog networks place work from instantaneous local readings, so migrations create hotspots, recovery waits on cluster leaders, and what one cluster learns is never reused by another.

SWARM-Fog treats scheduling, balancing and recovery as one problem: clusters share a fading "pheromone" map of what worked.

- No fixed controller: each cluster has a temporary, replaceable coordinator that cannot seize permanent control; scheduling is trust-aware and recovery plans are prepared before a node fails.
- Evaluation: on a controlled testbed (software-emulated enclaves), recovered fully across a wide range of failure rates, faster than four re-implemented baselines at equal completeness.

Also in preparation: MoCA (containing AI agents after a successful prompt injection), Fed-RAG-Hunt (sharing zero-day intelligence as differentially private summaries), a threshold-gated encrypted SIEM (whose risk tiers must not reveal which data is valuable) and FRESCO (replicas proven fresh before serving).

Leadership & Teaching

President – *Sub Student Committee 32, SIIT* 2024 – Present

Leads the committee that represents the SIIT student body and organises its academic and social calendar.

- Organised 11 student events and led SIIT First Meet 2024, welcoming about 450 first-year students with about 200 organisers.

Founder & Lead Instructor – *TOI Tutor and ABIG Center, online* 2022 – Present

After winning the national title, founded two online platforms that prepare students for the national olympiads (TOI, POSN).

- Taught 500+ hours to 50+ students; authored COM300 (C++ and algorithms, 128 lessons) and COM100 (maths and Python).

Skills & Interests

Security & IAM:	Zero Trust, OIDC/JWT, MFA (TOTP), RBAC/ABAC, OPA/Rego, Kubernetes authorisation
Governance & Risk:	ISO 27001, NIST, third-party risk management, due-diligence questionnaires, gap analysis
Cloud & DevOps:	AWS (EKS, EC2), Kubernetes, Docker, CI security scans (Semgrep, gitleaks), Azure basics
Cryptography:	zk-SNARKs (Groth16), Merkle trees, ML-DSA, ML-KEM, homomorphic encryption
AI & AI Security:	LLM agents, RAG, Model Context Protocol (MCP), prompt engineering, prompt-injection testing
Offensive & SOC:	Nmap, Metasploit, Burp Suite, Wireshark, pwntools, hashcat, OWASP Top 10; Splunk, Wazuh
Programming:	Python and C++ (main); Go, Rust, TypeScript, C, Java, Bash, SQL
Languages:	Thai (native), English (fluent), Chinese (basic)
Interests:	home-lab security, customising Linux setups, Muay Thai, board games, food and travel

Awards, Certifications & Training

- **1st in Thailand**, Thailand Olympiad in Informatics (TOI18): national champion and gold medallist, 2022.
- **Best Paper Award**, IEEE JCSSE 2026 (Joint Conference on Computer Science and Software Engineering): U-HAP, second author.
- **1st place (team)**, Image Recognition Programming Challenge, Osaka Institute of Technology and SIIT, 2025.
- **ISC2 Certified in Cybersecurity (CC)**, 2025 · CompTIA Security+ (exam planned Dec 2026).
- **Hands-on practice:** NCSA CTF Thailand Tournament 2026 (team), Hack The Scammer CTF 2025, TryHackMe, Hack The Box.
- **Labs:** SIIT Ethical Hacking and Digital Forensics (2026), SOSECURE Road to Cybersecurity (2025), Kali home lab.

References

- **Apichai Wongchindakhun**, CISO, Allianz Ayudhya; internship mentor | linkedin.com/in/apichai-wongchindakhun-1839b314
- **Prof. Somchart Fugkeaw**, SIIT, Thammasat University; research supervisor | linkedin.com/in/somchart-fugkeaw-a147b9111
- **Pachara Wongthanakarn**, TU Alert co-developer | linkedin.com/in/pachara-wongthanakarn